EXPERIMENT-7

BASIC GATES USING UNIVERSAL GATES

AIM: Implement basic gates using universal gates

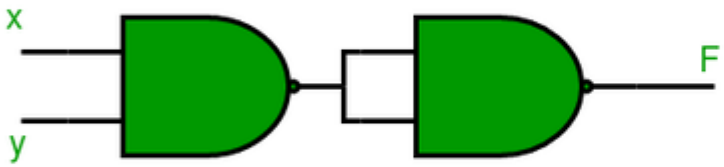
Components required : IC 7400,7402,Trainer kit

Theory: In Boolean Algebra, the **NAND** and **NOR** gates are called **universal gates** because any digital circuit can be implemented by using any one of these two i.e. any logic gate can be created using NAND or NOR gates only.

1. Implementation of AND Gate using Universal gates.

a) Using NAND Gates

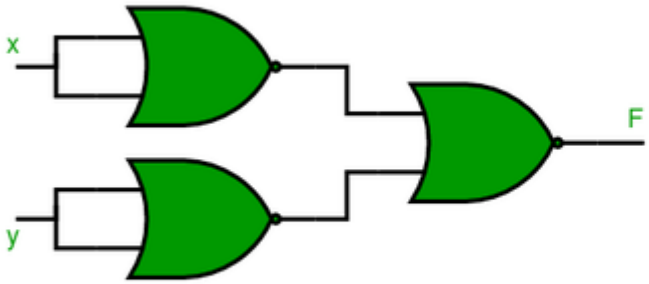
The AND gate can be implemented by using two NAND gates in the below fashion:



x	y	F
0	0	0
0	1	0
1	0	0
1	1	1

b) Using NOR Gates

Implementation of AND gate using only NOR gates as shown below:

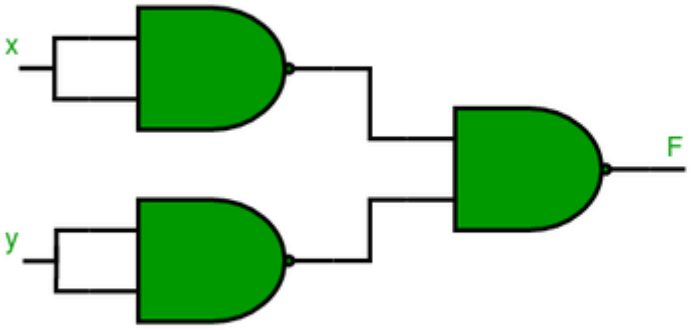


x	y	F
0	0	0
1	0	0
1	1	1
0	1	0

Implementation of OR Gate using Universal gates.

a) Using NAND Gates

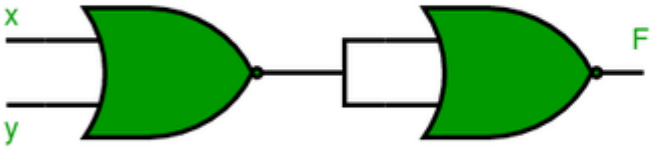
The OR gate can be implemented using the NAND gate as below:



x	y	F
0	0	0
0	1	1
1	0	1
1	1	1

b) Using NOR Gates

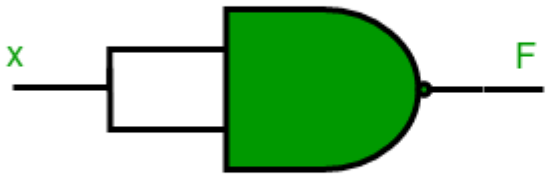
Implementation of OR gate using two NOR gates as shown in the picture below:



x	y	F
0	0	0
0	1	1
1	0	1
1	1	1

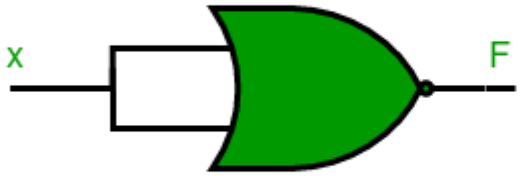
3. Implementation of NOT Gate using Universal gates.

a) Using NAND Gates



x	F
0	1
1	0

b) Using NOR Gates



x	F
0	1
1	0